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An Interactive PDF Chatbot Using Semantic Embeddings and FAISS-Based Vector Search

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Introduction

- Digitization has made PDFs a common format for documents worldwide.
- However, their static structure makes it difficult to access information quickly.
- Traditional keyword searches often fail to capture the actual meaning of content.
- This limitation causes frustration for users trying to find precise answers.
- Advances in Natural Language Processing (NLP) — such as transformers, dense embeddings, and conversational LLMs — enable deeper semantic understanding.
- This project introduces a chatbot that allows users to upload a PDF and ask natural-language questions, receiving accurate and context-aware answers instantly.

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Problem Statements

- Manually reading through lengthy PDFs is time-consuming and prone to errors.
- Keyword-based search methods often miss synonyms or contextual meaning.
- This results in irrelevant or incomplete excerpts being retrieved.
- Many current tools handle retrieval and generation as separate tasks, causing fragmented or inaccurate (hallucinated) answers.
- Non-technical users often find existing systems difficult due to complex query syntax.
- There is currently no simple, unified system that converts any PDF into a conversational knowledge base without specialized training.

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Objectives

- Enable users to query any PDF using natural language through a simple web interface.
- Improve contextual accuracy by combining semantic embeddings with scalable vector search.
- Generate coherent and meaningful answers using an instruction-tuned generative model without additional fine-tuning.
- Ensure real-time performance even on standard, non-specialized hardware.
- Offer an end-to-end solution that is easy to use and accessible for non-technical users.
- Maintain transparency and reproducibility across the entire system pipeline.

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Literature Review

- Early chatbots were rule-based and lacked contextual understanding.
- The introduction of **BERT** enabled bidirectional context comprehension in language models.
- **Sentence-BERT (SBERT)** further improved performance by generating sentence-level embeddings for information retrieval.
- **FAISS** made it possible to perform large-scale similarity searches efficiently.
- **FLAN-T5** demonstrated that instruction tuning significantly enhances zero-shot text generation.
- Existing PDF assistants either focus on retrieval or generation, but none integrate semantic indexing, fast vector search, and generative answering into a single lightweight system.

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Methodology

- The system uses **PyMuPDF** to extract text efficiently from PDF documents.
- A **sliding-window chunking** technique is applied to preserve contextual flow across sections.
- all-MiniLM-L6-v2** model encodes these text chunks into **384-dimensional embeddings**.
- FAISS** builds an approximate nearest-neighbor index for fast similarity search.
- User queries are embedded, and the **top-k relevant chunks** are retrieved using cosine similarity, concatenated with the question, and sent to **FLAN-T5-XL** for concise answer generation.
- The complete pipeline operates on **CPU or GPU** hardware without requiring any additional fine-tuning or training.

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Result & Discussion

- The zero-shot system achieves **87.99% semantic similarity** with reference answers.
- It also attains **84% cosine overlap** and **90% top-3 retrieval accuracy**.
- The average response time is **3–5 seconds**, outperforming the baseline CoPilot system.
- Minor errors appear in ambiguous queries, but the generated answers remain coherent.
- Embedding heatmaps show strong **semantic clustering**, confirming the model's understanding.
- These results validate the pipeline's **retrieval quality** and **practical usability**.

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Conclusion & Future Scope

- The integration of **Sentence-BERT**, **FAISS**, and **FLAN-T5-XL** enables an accurate and real-time PDF-based chatbot without fine-tuning.
- The system democratizes access to static PDF documents through natural, conversational interaction.
- It provides an efficient and user-friendly way to retrieve meaningful information from large text sources.
- Future enhancements include **layout-aware parsing** and **multimodal support** for figures and visual content.
- Planned upgrades also involve **domain-specific fine-tuning**, **multilingual embeddings**, and **cloud API deployment**.

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